SwiftDrop customer AI - guarded tool-calling sequence

Authenticated CUSTOMER
CustomerAiAssistant
API gateway :8080
AiChatService :8089
ChatSessionStore - memory
LlmClient - mock or real provider
ToolExecutor and HTTP clients
Customer order service
Gateway public tracking
Pricing service
Analytics PostgreSQL
1 Ask about own parcel or delivery price
2 POST /api/v1/ai/chat - message, optional conversationId, Bearer token
3 Verify JWT and forward trusted identity and correlation ID
alt
[Missing or invalid token]
4 401
[Authenticated caller]
5 Forward chat request with caller context
alt
[Caller role is not CUSTOMER]
6 403
7 Customer account required
[CUSTOMER caller]
8 Validate message length - 1 to 2000 characters
9 Consume shared 10 requests per minute and acquire owned conversation
alt
[Rate limited or conversation unknown or foreign]
10 429 or account-safe 404
11 Safe HTTP error
12 Visible failure and retry guidance
[Conversation acquired]
13 Redacted history - 30-minute idle lifetime, at most 8 prior turns
14 Lock session, redact input, apply private-data guard
alt
[Request asks for private data or credentials]
15 Audit chat REFUSED
16 Persist metadata-only ai_tool_invocation_audit row
17 Audit saved
18 Refusal audited
19 Produce fixed safe refusal
[Allowed request]
loop
[Model turns until final answer - at most 3 tool executions total]
20 System rules, redacted history and prompt, prior safe tool results
alt
[Provider unavailable]
21 Provider failure
break
[Provider failure stops the request]
22 Audit PROVIDER_UNAVAILABLE
23 Persist safe audit metadata
24 503 - no silent provider fallback
25 Safe provider-unavailable error
[Model responds]
26 Final answer or tool calls with arguments
opt
[Model requests a tool and remaining call budget exists]
27 Execute allowed tool with caller token and correlation ID
28 Allowlist name and validate exact argument schema
alt
[Unknown tool or invalid arguments]
29 REFUSED or INVALID_ARGUMENTS - no downstream HTTP
[track_parcel]
30 GET /api/v1/orders/tracking/{number} - caller Bearer token
31 Protected order lookup
32 Owned order or access error
33 Protected response - never passed to model
34 Require returned customerId to match caller userId
alt
[Unknown parcel or another customer's parcel]
35 Same NOT_FOUND outcome, do not call public tracking
[Ownership verified]
36 GET /api/v1/public/track/{number}
37 Read sanitized tracking view
38 Public tracking fields
39 Public view
40 Allowlist status, tracking number and districts only
[calculate_estimate]
41 POST /api/v1/pricing/estimate - weight, districts, STANDARD speed
42 Forward authenticated request
43 ApiResponse - estimatedFee and currency
44 Real pricing result
45 Validate nonnegative LKR fee and select safe estimate fields
46 Save audit - safe name, actor, role, correlation, outcome and duration
alt
[Audit persistence fails]
47 Persistence failure
break
[Audit failure stops the request]
48 Fail closed - no tool result released
49 500
50 Safe request failure
[Audit persisted]
51 Audit saved
52 Safe tool result and outcome
alt
[Outcome is REFUSED]
break
[Stop the model loop]
53 Produce fixed allowed-tools-only reply
[Validated execution outcome]
54 Append safe tool result for the next model turn
opt
[Requested tools exceed the budget]
55 Audit LIMIT_REACHED and stop further executions
56 Persist metadata-only audit
57 Produce fixed call-limit reply
opt
[Chat completes successfully]
58 Redact final reply
59 Retain redacted user and assistant turns - memory only
60 ApiResponse - reply, conversationId and safe tool-use labels
61 200
62 Reply, tool outcomes and preserved conversation